Assignment 1

Program -

```
1]class Node
{
  constructor(element)
  {
    this. element=element;
    this.next=null;

  }
}
class LinkedList
{
 constructor()
 {
   this.head=null;
   this.size=0;

 }
 add(element)
 {
   var temp=new Node(element);
   var temp1;
   if(this.head==null)
   this.head=temp;
   else
   {
     temp1= this.head;
     while(temp1.next)
     {
       temp1=temp1.next
     }
     temp1.next=temp;

   }
   this.size++;
   }
   insertAt(element, index)
  {
  if(index>0 && index>this.size)
  return false;
  else
  {
  var temp=new Node(element);
  var temp1, temp2;
  temp1=this.head;
  if (index==0)
  {
    this.next=this.head;
    this.head=temp;

  }
  else
  {
    temp1=this.head;
```

```
    var it=0;
    while(it<index)
    {
      it++;
      temp2=temp1;
      temp1=temp1.next;
  }
    temp.next = temp1;
       temp2.next = temp;
  }
  this.size++;
  }
}
PrintList()
 {
   var temp1=this.head;
   while(temp1)
   {
     console.log(temp1.element);
     temp1=temp1.next;
   }
 }
findPrevious(item)
 {
   var currNode = this.head;
   while (!(currNode.next == null) &&(currNode.next.element != item))
   {
     currNode = currNode.next;
   }
   return currNode;
 }
 remove(item)
 {
   var prevNode = this.findPrevious(item);
   if (!(prevNode.next == null))
   {
     prevNode.next = prevNode.next.next;
   }
 }
}
 ll =new LinkedList();
 ll.add(10);
 ll.add(20);
 ll.add(30);
 ll.insertAt(50,1);
 console.log(ll.insertAt(100,6));
 ll.PrintList();
 console.log("after removing the given element, the result is :")
ll.remove(30);
 ll.PrintList();
```

```
Output -

false
10
50
20
30
```

```
after removing the given element, the result is :
10
50
20
```